Supplementary Material: When Retrieval Quality Decouples from Faithfulness

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A Artifact Manifest

The double-blind supplement includes repository-local source tables and result files. Public links are anonymized for review; code and data will be released upon acceptance.

Table 1: Main evidence sources used by the paper. Paths are repository-local inside the anonymized artifact.

Claim area	Source files
Matched CCS test	`results/revision/fix_01/paired_wilcoxon.csv`
Scaled pilot cell	`results/revision/fix_02/headline_table.csv`
Multi-metric scoring	`results/revision/fix_03/table1_multimetric.csv`
Human calibration	`data/revision/fix_03/human_eval_*.csv`
Threshold transfer	`results/revision/fix_04/tau_transfer_matrix.csv`
Noise separation	`results/revision/fix_05/slope_response.csv`
Full Self-RAG audit	`data/revision/fix_06/per_query_full_selfrag.csv`
Baseline summary	`results/revision/fix_06/h2h_summary_full_selfrag.csv`
RAPTOR cost table	`results/revision/fix_11/raptor_full_table.csv`

B Human-Evaluation Details

The human calibration uses 99 examples sampled from the fixed-generation multi-metric cell. Two raters label each answer as `supported`, `partially_supported`, or `unsupported` using only the provided retrieved passages. Disagreements are resolved in an adjudicated CSV.

Table 2: Human calibration label distribution after adjudication.

Label	Count	Proportion
Supported	75	0.758
Partially supported	20	0.202
Unsupported	4	0.040

Raw agreement is 0.919 and Cohen's $\kappa = 0.774$. The correlation table in the main paper is computed from adjudicated ordinal labels (`supported` $= 1$, `partially_supported` $= 0.5$, `unsupported` $= 0$). Kendall τ_b values are 0.114 for DeBERTa, 0.311 for the second NLI scorer, and 0.421 for the RAGAS-style judge.

Annotation instructions. Raters judge answer support by the retrieved passages only. Outside knowledge is disallowed. If the context supports the main answer but the answer adds an unsupported qualifier or side detail, raters use `partially_supported`. If the answer says the information is unavailable, raters mark it supported only when the provided context really omits the answer.

C Self-RAG Artifact Traceability

The full Self-RAG package has been imported into repository-local files, so the paper does not depend on a path outside the anonymized artifact. The verified import contains the expected 1,600 per-query rows and 8 summary rows.

Table 3: Full Self-RAG artifact checksums.

File	Rows	SHA-256
`data/revision/fix_06/per_query_full_selfrag.csv`	1600	`aaf00af2f186a3d90c619c3885df2627e237580c54ac5837be31443f1360a5af`
`results/revision/fix_06/h2h_summary_full_selfrag.csv`	8	`5d325428d24c7d62b4fa22e30108edec1f5842d862536fc75b5f56192dc82055`
`results/revision/fix_06/summary_full_selfrag.md`	–	`cba4cadcd97b2705c6aaef7893816e7d2bdc1ce8f2ee67d92dce1fe537ba1574`

C.1 Harness-Mismatched Baseline: Self-RAG

Self-RAG is included for artifact traceability, but it is not a like-for-like matched-harness baseline. The run uses the released fine-tuned Llama-2-7B Self-RAG checkpoint with 8-bit loading, while CRAG, HCPC-v2, and RAPTOR-2L use the Mistral-backed short-answer harness. Self-RAG is also trained for reflective decoding, whereas the entailment-style scorer used in the audit does not separately reward reflection tokens or task-specific citation behavior. These numbers should therefore be read as harness-conditional, not as a global comparison against Self-RAG.

Table 4: Harness-mismatched Self-RAG baseline. Latency is milliseconds.

Dataset	Faith	Halluc.	Mean lat.	p99 lat.
SQuAD	0.574	0.390	7542	44800
HotpotQA	0.574	0.430	6158	46965

D Reproducibility Commands

All revision runs are designed for free local or Kaggle compute. The exact runbook is included in `docs/revision/runbook.md`. Representative commands are:

```
python3 experiments/fix_01_causal_matched_pairs.py
python3 experiments/fix_02_scaled_headline_n500.py
python3 experiments/fix_03_multimetric_faithfulness.py
python3 experiments/analyze_fix03_human_eval.py
python3 experiments/fix_04_tau_generalization.py
python3 experiments/fix_05_coherence_preserving_noise.py
python3 experiments/fix_06_baseline_h2h_pareto.py
python3 experiments/fix_11_raptor_full_table.py
```

E Anonymous Release Placeholders

During double-blind review, public author-identifying links are replaced with placeholders such as [anonymized repository], [anonymized dataset], and [anonymized DOI]. Post-review sources can restore the public repository, dataset, package, and DOI links after review.